Q1. What is DevOps?

A. DevOps is a culture and set of practices that combine software development and IT operations to improve collaboration, automation, and delivery speed.

Q2. What is CI/CD?

A. CI/CD stands for Continuous Integration and Continuous Delivery/Deployment. It automates code integration, testing, and deployment.

Q3. Infrastructure as Code?

A. Infrastructure as Code (IaC) manages infrastructure through code and configuration files instead of manual processes.

Q4. Terraform?

A. Terraform is an open-source IaC tool used to provision and manage infrastructure across multiple cloud providers.

Q5. Ansible?

A. Ansible is an automation tool used for configuration management, application deployment, and orchestration.

Q6. Docker?

A. Docker is a platform that packages applications and dependencies into portable containers.

Q7. Kubernetes?

A. Kubernetes is an orchestration platform that automates deployment, scaling, and management of containerized applications.

Q8. What is a container?

A. A container is a lightweight, portable unit that includes an application and all its dependencies.

Q9. VM vs container?

A. Virtual Machines include a full operating system, while containers share the host OS kernel and are more lightweight.

Q10. What is AWS?

A. Amazon Web Services (AWS) is a cloud computing platform offering infrastructure, storage, databases, and AI services.

Q11. What is Azure?

A. Microsoft Azure is a cloud platform that provides computing, storage, networking, and enterprise services.

Q12. What is GCP?

A. Google Cloud Platform (GCP) is Google's cloud computing service offering infrastructure and managed solutions.

Q13. EC2?

A. Amazon EC2 provides scalable virtual servers in the AWS cloud.

Q14. S3?

A. Amazon S3 is an object storage service designed for durability, scalability, and security.

Q15. What is VPC?

A. A Virtual Private Cloud (VPC) is a logically isolated network within a cloud environment.

Q16. Load balancing?

A. Load balancing distributes traffic across multiple servers to improve performance and availability.

Q17. Auto scaling?

A. Auto scaling automatically adjusts computing resources based on workload demand.

Q18. DNS?

A. Domain Name System (DNS) translates domain names into IP addresses.

Q19. CDN?

A. A Content Delivery Network (CDN) caches content across geographically distributed servers to reduce latency.

Q20. Nginx?

A. Nginx is a high-performance web server, reverse proxy, and load balancer.

Q21. Apache web server?

A. Apache HTTP Server is a widely used open-source web server for hosting websites and applications.

Q22. Linux permissions?

A. Linux permissions control access to files and directories using read, write, and execute privileges.

Q23. Bash scripting?

A. Bash scripting automates repetitive tasks and system administration activities in Linux.

Q24. Git?

A. Git is a distributed version control system used to track code changes and collaborate on software projects.

Q25. GitHub Actions?

A. GitHub Actions automates workflows such as building, testing, and deploying applications.

Q26. Jenkins?

A. Jenkins is an open-source automation server commonly used for CI/CD pipelines.

Q27. Monitoring?

A. Monitoring tracks system health, performance, and availability using metrics and alerts.

Q28. Prometheus?

A. Prometheus is an open-source monitoring system that collects and stores time-series metrics.

Q29. Grafana?

A. Grafana is a visualization platform used to create dashboards and analyze monitoring data.

Q30. Logging?

A. Logging is the process of recording system events and application activity for troubleshooting and auditing.

Q31. ELK Stack?

A. The ELK Stack consists of Elasticsearch, Logstash, and Kibana for centralized log management and analysis.

Q32. Observability?

A. Observability is the ability to understand system behavior through metrics, logs, and traces.

Q33. Blue-green deployment?

A. Blue-green deployment uses two identical environments to minimize downtime during releases.

Q34. Canary deployment?

A. Canary deployment gradually rolls out changes to a small subset of users before full release.

Q35. Rolling deployment?

A. Rolling deployment updates instances incrementally without taking the entire service offline.

Q36. Microservices?

A. Microservices architecture breaks applications into small, independent services.

Q37. Service mesh?

A. A service mesh manages communication, security, and observability between microservices.

Q38. Kubernetes ingress?

A. Ingress manages external access to services within a Kubernetes cluster.

Q39. Secrets management?

A. Secrets management securely stores and controls access to sensitive credentials and keys.

Q40. HashiCorp Vault?

A. HashiCorp Vault is a tool for managing secrets, encryption keys, and secure access.

Q41. SSL/TLS?

A. SSL/TLS protocols encrypt data in transit and secure network communications.

Q42. High availability?

A. High availability ensures systems remain operational with minimal downtime.

Q43. Disaster recovery?

A. Disaster recovery involves restoring systems and services after failures or catastrophic events.

Q44. Fault tolerance?

A. Fault tolerance enables systems to continue operating even when components fail.

Q45. Network troubleshooting?

A. Network troubleshooting identifies and resolves connectivity, routing, DNS, and performance issues.

Q46. Production outage example

A. Explain the issue, investigation process, root cause, resolution, and lessons learned.

Q47. Optimize cloud costs

A. Optimize costs through right-sizing, auto-scaling, reserved instances, storage lifecycle policies, and monitoring.

Q48. Deployment process

A. Describe code commit, CI pipeline, testing, approval, deployment, monitoring, and rollback procedures.

Q49. Metrics monitored

A. Common metrics include CPU, memory, disk usage, network traffic, latency, error rates, and availability.

Q50. Questions for us?

A. Ask about team practices, cloud architecture, deployment frequency, challenges, and growth opportunities.